

WEEK 11

Task 1

Skill objective:

To Implement how to List Active Jobs, Display Status Information, Retrieve Job IDs, Format Output, Update State Information, Test Job Listings.

Implementation / work:

```
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

typedef enum { RUNNING, STOPPED, DONE } Status;

typedef struct {
    int jid;    pid_t
    pid;    Status
    status;    char
    cmd[40]; }
JobRecord;

void list_jobs(JobRecord jobs[], int n) {    printf("JID    PID    STATUS
COMMAND\n");    printf("-----\n");
    for (int i = 0; i < n; i++) {        const char *stat_str = (jobs[i].status ==
RUNNING) ? "Running" :        (jobs[i].status == STOPPED)
? "Stopped" : "Done";        printf("[%d]    %-7d    %-10s    %s\n",
        jobs[i].jid, jobs[i].pid, stat_str, jobs[i].cmd);
    } }

int main() {    JobRecord
active_jobs[] = {
    {1, 2301, RUNNING, "sleep 100 &"},
    {2, 2305, STOPPED, "vim server.conf"},
    {3, 2312, RUNNING, "find / -name '*.log' > logs.txt &"},
    {4, 2318, DONE, "tar -czf backup.tar.gz src/" }
};
printf("[Task 1] Listing active shell jobs and formatted statuses:\n\n");
```

Skill objective:

```
list_jobs(active_jobs, 4);    return 0; }
```

Output:

```
naveen123@LAPTOP-4NVQJ2C7:~$ nano exp11s.c
naveen123@LAPTOP-4NVQJ2C7:~$ gcc exp11s.c -o exp11s
naveen123@LAPTOP-4NVQJ2C7:~$ ./exp11s
[Task 1] Listing active shell jobs and formatted statuses:

JID PID STATUS COMMAND
-----
[1] 2301      Running  sleep 100 &
[2] 2305      Stopped  vim server.conf
[3] 2312      Running  find / -name '*.log' > logs.txt &
[4] 2318      Done     tar -czf backup.tar.gz src/
naveen123@LAPTOP-4NVQJ2C7:~$ |
```

Task 2

To design how to Identify Target Job, Transfer Terminal Control, Resume Processes, Wait for Completion, Update Job State, Test Foreground Switching.

Implementation / work:

```
#include <stdio.h>
#include <signal.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {    printf("[Task 2] Foreground Switching Simulation (fg %%1)\n");    pid_t pid =
fork();    if (pid == 0) {        /* Child simulated process */        usleep(400000);
printf("[Job 1] Child process (PID %d) resuming foreground execution...\n", getpid());
usleep(300000);        _exit(0);    }

    printf("1. Target Job identified: JID [1] -> PID: %d\n", pid);
printf("2. Sending SIGCONT signal to resume stopped/background job...\n");
kill(pid, SIGCONT);

    printf("3. Transferring terminal control to PID %d (tcsetpgrp simulation)\n", pid);
int status;    waitpid(pid, &status, WUNTRACED);
```

Skill objective:

```
printf("4. Process completed. Terminal ownership restored to shell.\n");
printf("5. Job state updated in Job Table: JID [1] -> TERMINATED.\n");
return 0; } Output:
```

```
naveen123@LAPTOP-4NVQJ2C7:~$ nano exp11s2.c
naveen123@LAPTOP-4NVQJ2C7:~$ gcc exp11s2.c -o exp11s2
naveen123@LAPTOP-4NVQJ2C7:~$ ./exp11s2
[Task 2] Foreground Switching Simulation (fg %1)
1. Target Job identified: JID [1] -> PID: 1678
2. Sending SIGCONT signal to resume stopped/background job...
3. Transferring terminal control to PID 1678 (tcsetpgrp simulation)
[Job 1] Child process (PID 1678) resuming foreground execution...
4. Process completed. Terminal ownership restored to shell.
5. Job state updated in Job Table: JID [1] -> TERMINATED.
```

Task 3

To Analyze Signal Delivery, Understand Signal Masks, Review Process Groups, Explore Terminal Signals, Test Signal Behavior.

Implementation / work:

```
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void analyze_signal_mask() {
    sigset_t
    old_mask, block_mask;
    sigemptyset(&block_mask);
    sigaddset(&block_mask, SIGINT); /* Ctrl+C */
    sigaddset(&block_mask, SIGTSTP); /* Ctrl+Z */

    printf("[Task 3] Blocking SIGINT and SIGTSTP via sigprocmask(SIG_BLOCK)...\n");
    sigprocmask(SIG_BLOCK, &block_mask, &old_mask);

    printf("Critical section active: Shell data structures safely protected.\n");
    printf("Process Group ID: %d | Foreground Group: %d\n", getpgrp(), tcgetpgrp(STDIN_FILENO));

    printf("Restoring signal mask via sigprocmask(SIG_SETMASK)...\n");
    sigprocmask(SIG_SETMASK, &old_mask, NULL); printf("Signal delivery
    unblocked successfully.\n"); }

int main() {
    analyze_signal_mask();
    return 0; } Output:
```

Skill objective:

```
naveen123@LAPTOP-4NVQJ2C7:~$ nano exp11s3.c
naveen123@LAPTOP-4NVQJ2C7:~$ gcc exp11s3.c -o exp11s3
naveen123@LAPTOP-4NVQJ2C7:~$ ./exp11s3
[Task 3] Blocking SIGINT and SIGTSTP via sigprocmask(SIG_BLOCK)...
Critical section active: Shell data structures safely protected.
Process Group ID: 1712 | Foreground Group: 1712
Restoring signal mask via sigprocmask(SIG_SETMASK)...
Signal delivery unblocked successfully.
```

Task 4

To implement how to resume Stopped Jobs, Send Continue Signals, Update Job Status, Maintain Job Table, Verify Execution, Test Background Recovery.

Implementation / work:

```
#include <stdio.h>
#include <signal.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    printf("[Task 4] Resume Stopped Jobs in Background (bg %1)\n");
    pid_t pid = fork();
    if (pid == 0) {
        usleep(300000);
        printf("[Background Worker PID %d] Processing asynchronous tasks...\n", getpid());
        _exit(0);
    }

    printf("Sending SIGCONT to stopped job PID: %d...\n", pid);
    kill(pid, SIGCONT);

    printf("Updating Job Table status: [1] RUNNING in background.\n");
    printf("Shell prompt returned immediately: teja27@TEJA:~$ \n");

    waitpid(pid, NULL, 0);
    printf("Background job PID %d completed successfully.\n", pid);
    return 0; }
```

Output:

Skill objective:

```
naveen123@LAPTOP-4NVQJ2C7:~$ nano exp11s4.c
naveen123@LAPTOP-4NVQJ2C7:~$ gcc exp11s4.c -o exp11s4
naveen123@LAPTOP-4NVQJ2C7:~$ ./exp11s4
[Task 4] Resume Stopped Jobs in Background (bg %1)
Sending SIGCONT to stopped job PID: 1749...
Updating Job Table status: [1] RUNNING in background.
Shell prompt returned immediately: naveen123@LAPTOP-4NVQJ2C7:~$
[Background Worker PID 1749] Processing asynchronous tasks...
Background job PID 1749 completed successfully.
naveen123@LAPTOP-4NVQJ2C7:~$
```